
Deep Reinforcement Learning Finds Bayes–Nash Equilibrium in Competitive Newsvendor Problems

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Abstract

We investigate learning dynamics in competitive newsvendor games, a class of continuous-action games with strategic substitutes. Despite established equilibrium properties, convergence of independent learning algorithms in repeated general-sum play remains uncertain. We analyze structural properties under complete and incomplete information, deriving closed-form equilibria for a symmetric complete-information benchmark with perfect substitution. Our main theoretical contribution proves strict monotonicity in both complete-information and Bayesian models with private costs, ensuring equilibrium uniqueness. This provides convergence guarantees for variational-inequality-based algorithms. Numerical experiments using deep reinforcement learning agents with Proximal Policy Optimization empirically demonstrate convergence to Nash and Bayesian Nash equilibria, verified by equilibrium checks. These results establish a foundation for applying deep reinforcement learning in competitive inventory management.

1. Introduction

Learning-based decision making increasingly takes place in *competitive multi-agent environments*, where autonomous agents interact repeatedly while adapting their strategies. Prominent examples include online marketplaces, digital advertising, pricing platforms, and supply-chain coordination. In such environments, agents do not face a stationary

data-generating process. Instead, the observations of each agent—prices, quantities, or realized demand—are *endogenous* and shaped by the simultaneous learning dynamics of other strategic agents.

This endogeneity fundamentally distinguishes learning in games from classical single-agent online learning or reinforcement learning (Fudenberg & Levine, 1999). While single-agent learners infer an exogenous environment, multi-agent learners collectively generate it. As a result, learning dynamics in general-sum games may converge to equilibrium, but they may also cycle, diverge, or exhibit chaotic behavior (Sanders et al., 2018). Understanding *when* and *why* learning converges in such settings remains a central challenge in multi-agent reinforcement learning.

In this paper, we study learning dynamics in a class of continuous-action general-sum games through the lens of a competitive newsvendor model (Parlar, 1988; Güler et al., 2018). While originating from inventory management, this serves as a canonical example of a continuous, non-potential game with strategic interaction and uncertainty, where agents choose real-valued actions under stochastic demand with substitution effects (a formal proof of non-potentiality is provided in Appendix A).

A key feature is that actions are *strategic substitutes*: increasing one agent’s action reduces the marginal incentive of others to increase theirs. Such games typically exhibit instability and non-convergence (Hinloopen, 2015). Our central observation is that the competitive newsvendor game with uniform demand constitutes an important exception: it satisfies *strict monotonicity* of the payoff gradient, which rules out cyclic behavior and provides convergence guarantees for equilibrium learning.

Understanding such structure is particularly important given the growing use of deep reinforcement learning (DRL) for decentralized decision making. DRL has been widely applied to inventory and supply-chain problems (Oroojlooyjavid et al., 2020; Gijsbrechts et al., 2022; Rolf et al., 2022; Dehaybe et al., 2024), but existing work largely treats these settings as single-agent control problems or relies on empirical convergence without structural explanations. In contrast, we explicitly connect the convergence behavior of learning algorithms to the underlying geometry of the game.

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We analyze both a complete-information setting, in which agents’ cost parameters are fixed, and a Bayesian setting with private cost information. The complete-information model admits a closed-form symmetric Nash equilibrium and serves as a transparent baseline for studying learning dynamics. The Bayesian model captures environments in which agents face heterogeneous and privately known costs, and equilibrium strategies are functions over finite type spaces. These two settings correspond naturally to repeated interaction among fixed competitors and to environments with changing participants.

Existing work characterizes equilibria in both settings but typically relies on implicit or numerical representations. More importantly, prior analyses do not explain why independent learning algorithms should converge in these games. This paper makes three contributions.

First, we show that the competitive newsvendor game with uniformly distributed demand is *strictly monotone*. We establish this property for the complete-information game with two agents and extend it to the Bayesian setting with private costs. Strict monotonicity implies equilibrium uniqueness and excludes cyclic or unstable learning dynamics. From a learning perspective, it places the game in a class where first-order methods for variational inequalities admit global convergence guarantees.

Second, we derive a closed-form characterization of the symmetric Nash equilibrium in the complete-information model. While not essential for learning, this analytical benchmark provides economic intuition and a concrete target for evaluating learning dynamics in continuous action spaces.

Third, we empirically study equilibrium learning using deep reinforcement learning. Using Proximal Policy Optimization (PPO) in self-play, we find that independently trained agents reliably converge to Nash and Bayes–Nash equilibria across symmetric and asymmetric environments. Convergence is certified via equilibrium verification. PPO leverages the structure of the game, and we find equilibrium with uniform but also with Gaussian demand.

It is interesting to observe the reported failures of convergence of reinforcement learning in Bertrand oligopoly and auction models (Calvano et al., 2020; Deng et al., 2024; Bichler et al., 2025). These models do not satisfy the monotonicity property (Bichler et al., 2025). Our work highlights monotonicity as a key property linking equilibrium theory and learning dynamics in continuous multi-agent environments.

2. Related Literature

In what follows, we review related work on competitive newsvendor problems, dynamic inventory models, and equi-

librium learning in continuous games.

2.1. Newsvendor Models

The newsvendor problem is one of the central models in inventory management. Its origins date back to Edgeworth (1888), while the modern formulation is commonly attributed to Arrow et al. (1951). Since then, it has inspired a vast literature in operations management and economics (Muller, 2019).

More recently, machine learning and deep reinforcement learning (DRL) has become a popular adaptive approach for solving inventory control problems in practice. For example, Gijbrecchts et al. (2022) show that DRL can serve as a powerful general-purpose technology that matches or slightly outperforms state-of-the-art heuristics without relying on problem-specific structural knowledge. Similarly, Dehaybe et al. (2024) demonstrate that DRL agents can learn flexible, dynamic inventory policies that adapt to evolving demand forecasts. These results highlight the strength of DRL in complex and non-stationary environments where classical analytical models or dynamic programming approaches become intractable. Importantly, this line of work models the newsvendor problem as a *single-agent* decision problem.

In contrast, the competitive newsvendor problem studies supply chains in which multiple decision-makers interact strategically under stochastic demand (de Kok & Graves, 2003). Early work analyzed competition under various demand-allocation schemes. Parlar (1988) studied a single-stage game with two firms managing substitutable products and independent demands, where a deterministic fraction of unsatisfied demand spills over to the competitor. They established the existence of a unique Nash equilibrium and showed that total profits are strictly lower than under cooperation. Lippman & McCardle (1997) proposed an alternative framework in which an aggregate industry demand is allocated across firms according to prespecified rules, with firms penalized for shortages both in their initial allocation and in any diverted demand. They likewise proved equilibrium existence and uniqueness.

These models were later extended to n retailers by Netessine & Rudi (2003) and Wang & Parlar (1994), who focused primarily on equilibrium existence and comparative statics of inventory levels. A common limitation of this early literature is the assumption of symmetry and complete information. Addressing this gap, Güler et al. (2018) extended the models of Parlar (1988) and Lippman & McCardle (1997) to settings with private cost information. They show that private costs combined with spillover demand induce systematic overstocking relative to full-information benchmarks, with inefficiencies ranging from 0.16% to 1.2% depending on cost structures. This inefficiency arises from firms’ strategic incentives to capture diverted demand from

competitors.

2.2. Equilibrium Learning

Game-theoretical analyses for competitive newsvendor problems are analytical, which requires agents to possess a lot of information about others' costs. In practice, however, firms typically do not know demand distributions or competitors' characteristics *ex ante* and must learn them over time (Gijbrenchts et al., 2022). Our work, therefore, connects naturally to the literature on equilibrium learning, which studies which equilibria emerge when agents adapt their strategies through repeated interaction (Fudenberg & Levine, 1998).

Fictitious play (Brown, 1951) is a canonical learning process in which agents best respond to empirical frequencies of opponents' past actions. While fictitious play converges in some classes of games, it is well known that learning dynamics can fail to converge in general-sum games (Daskalakis et al., 2010). Instead, learning may cycle, diverge, or exhibit chaotic behavior (Mertikopoulos et al., 2018; Cheung & Piliouras, 2020). Only recently has convergence of learning dynamics been established for specific classes of stochastic and Bayesian games (Bichler et al., 2025; Ahunbay & Bichler, 2025; Sayin et al., 2022).

We are particularly interested in equilibrium computation and learning in Bayesian games with continuous types and actions. For smooth and concave Bayesian games, the Bayes–Nash equilibrium (BNE) can be characterized as the unique solution to a variational inequality defined on a Hilbert space of square-integrable strategy functions (Ui, 2016). This formulation provides a powerful link between equilibrium theory and learning dynamics, as properties of the game directly affect equilibrium uniqueness and convergence of some learning algorithms.

Recent work on learning in games has employed neural networks and self-play to approximate equilibria even in Bayesian game models (Bichler et al., 2021; 2023). For dynamic environments, Pieroth et al. (2025) propose a deep reinforcement learning approach for multi-stage Bayesian games that verifies a Bayes–Nash equilibrium *ex post*. We build on these methodological advances to study equilibrium learning in competitive newsvendor models, which were previously considered very challenging analytically and for which we do not know if learning algorithms converge.

3. The Competitive Newsvendor Model

This section introduces a formal model of strategic inventory competition between automated agents in supply chain networks. Building on the seminal framework of Parlar (1988), we model a competitive newsvendor game in which vendors choose inventory levels under demand uncertainty while accounting for demand substitution effects.

3.1. Setup and Market Structure

We consider a duopoly market in which two competing vendors $i \in \{1, 2\}$ simultaneously choose inventory levels y_i at the beginning of each selling period, before demand uncertainty is resolved. Inventory choices are restricted to a nonempty, convex, and compact action space $K_i \subset \mathbb{R}_0^+$, reflecting physical capacity constraints and ensuring well-posedness of the equilibrium and learning problems.

Each vendor faces an independent stochastic demand x_i with probability density function f_i and cumulative distribution function F_i . We assume that f_i is continuous and supported on a compact interval $[0, \bar{x}_i]$, with $\bar{x}_i < \infty$. When agent i is clear, we refer to its density by f and by j 's density by g to reduce notational clutter.

Strategic interaction arises through demand spillovers when vendors stock out. If vendor i cannot satisfy a unit of demand due to insufficient inventory, a fraction $a \in [0, 1]$ of this unmet demand transfers to vendor j , provided vendor j has available capacity. The substitution parameter a allows the model to interpolate between independent newsvendor problems ($a = 0$) and fully competitive environments ($a = 1$). Our theoretical analysis focuses on the economically most competitive case $a = 1$, which captures perfect substitution and yields the strongest strategic interaction. Formally, the total surplus demand from vendor j faced by vendor i is

$$d_i = a(x_j - y_j)^+,$$

where $(\cdot)^+$ denotes the positive part. We define the joint inventory vector $\mathbf{y} = (y_1, y_2) \in K_1 \times K_2 =: K$. This notation carries over to the cost vector $\mathbf{c} = (c_1, c_2)$ and other vectors for later reference.

We focus on a duopoly market because it is the smallest setting that captures strategic competition while keeping the spillover mechanism transparent. With two firms, inventory decisions already interact through demand substitution. Extending the model to n firms would require additional assumptions on how unmet demand is distributed across multiple rivals, and such rules are not unique; see Netesine & Rudi (2003) and Wang & Parlar (1994). Since these choices add substantial analytical complexity with only a mild additional contribution to our main question, we restrict attention to the duopoly case.

3.2. Profit Structure and Economic Assumptions

Each vendor's per-period profit depends on four economic parameters: inventory cost c_i , sales price s , penalty cost p incurred for unmet original demand, and salvage value q for unsold inventory. Given inventory decisions $\mathbf{y}$ and demand

realizations, the realized profit of vendor i is

$$\pi_i(\mathbf{y}, c_i, \mathbf{x}) := -c_i y_i + s \cdot \min(y_i, x_i + d_i) - p \cdot (x_i - y_i)^+ + q \cdot (y_i - x_i - d_i)^+. \quad (1)$$

This formulation captures the fundamental trade-off in inventory management between understocking, which leads to lost sales and penalty costs, and overstocking, which incurs inventory costs with limited salvage recovery.

Vendors aim to maximize expected profit,

$$J_i(\mathbf{y}, c_i) = \mathbb{E}_{\mathbf{X}}[\pi_i(\mathbf{y}, c_i, \mathbf{X})], \quad (2)$$

where the expectation is taken over all demand realizations. Each player's payoff therefore depends on both their own inventory choice and their competitor's decision through the demand spillover mechanism. We formalize this interaction as a continuous normal-form game.

One explicit representation of J_i is

$$\begin{aligned} J_i(\mathbf{y}, c_i) = & (s + p) \int_0^{y_i} x_i f(x_i) dx_i \\ & + (s + p) y_i \int_{y_i}^{\infty} f(x_i) dx_i - p \mathbb{E}(X_i) \\ & + (s - q) \int_0^{y_i} \int_{y_j}^{B_i} d_i g(x_j) f(x_i) dx_j dx_i \\ & + (s - q) \int_0^{y_i} \int_{B_i}^{\infty} (y_i - x_i) g(x_j) f(x_i) dx_j dx_i \\ & + q \int_0^{y_i} (y_i - x_i) f(x_i) dx_i - c_i y_i, \end{aligned}$$

where $B_i := \frac{y_i - x_i}{a} + y_j$ denotes the spillover threshold and $d_i = a(x_j - y_j)^+$ (Parlar, 1988). The first and second term capture expected revenue from serving own demand when demand is below and above inventory; and the third captures expected penalty costs for unmet original demand. The fourth and fifth terms represent expected spillover revenues when the rival stocks out, distinguishing whether spillover demand is insufficient or sufficient to absorb player i 's remaining inventory. The sixth term captures expected salvage value of unsold inventory, and the final term accounts for inventory acquisition costs.

Under the maintained assumptions on demand distributions and action spaces, $J_i(\mathbf{y}, c_i)$ is continuously differentiable in y_i on the interior of K_i . We define the marginal profit function

$$I_i(\mathbf{y}, c_i) := \frac{\partial J_i(\mathbf{y}, c_i)}{\partial y_i}, \quad (3)$$

and collect marginal profits in the vector $\mathbf{I}(\mathbf{y}, \mathbf{c}) := (I_1(\mathbf{y}, c_1), I_2(\mathbf{y}, c_2))$. Explicitly, building on Equation (2)

the marginal profit function as shown by Parlar (1988) can be written as

$$I_i(\mathbf{y}, c_i) = (s + p) \int_{y_i}^{\infty} f(x_i) dx_i + q \int_0^{y_i} f(x_i) dx_i \quad (4)$$

$$+ (s - q) \int_0^{y_i} \int_{B_i}^{\infty} g(x_j) f(x_i) dx_j dx_i - c_i. \quad (5)$$

We impose several standard assumptions to ensure economic realism and analytical tractability. First, vendors begin each selling period with zero initial inventory. Second, we assume $s > c_i > q$, ensuring positive margins from sales and limited salvage recovery. Third, penalty costs apply only to a vendor's original demand shortfall $(x_i - y_i)^+$ and not to unmet spillover demand $(d_i + x_i - y_i)^+$.

3.2.1. BAYESIAN EXTENSION

In many practical scenarios, firms' costs vary across competitors and over time and are more naturally modeled as random variables. To capture this uncertainty, we extend the model to a Bayesian framework in which each vendor's cost parameter c_i is drawn from a finite type space $C_i \subset [0, s]$ according to a commonly known prior distribution μ_i . We assume independent types across players as in the standard independent private values model.

Each vendor observes their own realized cost type c_i , but not that of their competitor, when choosing inventory levels. A (pure) strategy for vendor i is therefore a mapping $y_i : C_i \rightarrow K_i$, which assigns an inventory level to each possible cost type. We denote a strategy profile by $\mathbf{y} = (y_1(\cdot), y_2(\cdot))$, and write $K_i := K_i^{[C_i]}$ for the corresponding strategy space, with $\mathcal{K} := K_1 \times K_2$.

Given a realized cost type c_i , the expected profit of vendor i under strategy profile $\mathbf{y}$ is

$$\hat{J}_i(\mathbf{y}, c_i) := \mathbb{E}_{c_j \sim \mu_j} [J_i(y_i(c_i), y_j(c_j), c_i)]. \quad (6)$$

For a given type c_i , we define the *type-wise marginal profit* as the derivative of $\hat{J}_i(y, c_i)$ with respect to the action chosen at that type:

$$\hat{I}_i(\mathbf{y}, c_i) := \frac{\partial}{\partial y_i(c_i)} \mathbb{E}_{c_j \sim \mu_j} [J_i(y_i(c_i), y_j(c_j), c_i)] \quad (7)$$

$$= \mathbb{E}_{c_j \sim \mu_j} [I_i(y_i(c_i), y_j(c_j), c_i)]. \quad (8)$$

4. Complete-Information Nash Equilibrium

In this section, we establish closed-form solutions for the Nash equilibrium of the symmetric competitive newsvendor game $\mathbf{G}$. These results serve as analytical benchmarks for the learning dynamics studied later. Throughout this section, we focus on symmetric environments in which the

cost parameter $c_1 = c_2$ is identical across players, and we therefore drop player indices.

To build intuition, we first analyze a simplified version $\mathbf{G}^*$ of the game with no penalty costs ($p = 0$), no salvage value ($q = 0$), and perfect substitution ($a = 1$). Our analysis proceeds in two stages. We first derive closed-form equilibrium solutions for this simplified setting, which allows for clear economic interpretation. We then extend the analysis to characterize the equilibrium structure under arbitrary parameter values.

While Parlar (1988) established existence and uniqueness of equilibrium, the equilibrium conditions are expressed as systems of integrals that generally require numerical solution. By solving these integrals explicitly for relevant parameter regimes, we obtain tractable closed-form equilibria that provide precise, closed-form targets for learning algorithms.

Theorem 4.1. *In the simplified symmetric competitive newsvendor game $\mathbf{G}^*$ with $p = 0$, $q = 0$, and $a = 1$, where demand is uniformly distributed on $[0, 1]$, the unique symmetric Nash equilibrium inventory level for both players is given by*

$$y_i = \begin{cases} 2 - \sqrt{1 + 2\frac{c}{s}}, & \text{if } \frac{c}{s} \leq \frac{5}{8}, \\ \sqrt{\frac{2}{3}} \left(1 - \frac{c}{s}\right), & \text{if } \frac{c}{s} > \frac{5}{8}. \end{cases} \quad (9)$$

The proof of this theorem is provided in Appendix B. The simplified model highlights two opposing strategic forces. On the one hand, firms face the standard newsvendor trade-off between overstocking and understocking. On the other hand, inventory choices are strategic substitutes: increasing one firm's inventory reduces the spillover demand available to its competitor, thereby lowering the competitor's marginal incentive to stock. The equilibrium structure reflects this interaction. When inventory costs are high ($c/s \geq 5/8$), firms optimally choose relatively low inventory levels and equilibrium arises in a low-stock regime. When costs are sufficiently low, competition for spillover demand induces firms to stock more aggressively, leading to a high-inventory equilibrium.

To accommodate economically relevant features such as penalty costs, salvage values, and imperfect substitution, we next consider the general parameter case.

Theorem 4.2. *Consider the symmetric competitive newsvendor game $\mathbf{G}$ with arbitrary parameters $(p, q, s, c, a) \in \mathbb{R}_+^5$ satisfying $s > q$ and $a > 0$, where demand is uniformly distributed on $[0, 1]$. The unique symmetric Nash equilibrium inventory level satisfies the following. If*

$$s + p - c \leq \frac{ap}{a+1} + \frac{a(2a+1)}{2(a+1)^2}(s-q), \quad (10)$$

then

$$y_i = \frac{-ap + \sqrt{(ap)^2 + 2a(s+p-c)(s-q)(2a+1)}}{(s-q)(2a+1)}.$$

Otherwise,

$$y_i = 1 + \frac{1}{a} + \frac{p}{a(s-q)} - \frac{\sqrt{(p+s-q)^2 + 2a(s-q)(c-q)}}{a(s-q)}.$$

The proof is again provided in Appendix B. The equilibrium characterization in Theorem 4.2 again reflects a regime switch driven by the interaction of inventory costs, demand spillovers, and penalty and salvage parameters. The inequality in the condition separates parameter regions in which equilibrium inventory is determined by interior marginal conditions from regions in which firms optimally operate at higher aggregate inventory levels to capture spillover demand. While the explicit expressions are algebraically involved, they provide closed-form predictions that serve as precise analytical benchmarks for the learning dynamics studied later.

We maintain the assumption of uniform demand in Sections 5 and 6. This choice has two advantages. First, it yields closed-form expressions for the relevant integrals, making the equilibrium and monotonicity analysis tractable. Second, the uniform distribution serves as a neutral benchmark: its symmetry avoids introducing additional distributional features, such as skewness or tail effects, that could obscure the strategic effects of substitution and spillovers.

5. Bayes-Nash Equilibrium

The simplifying assumptions of symmetry, uniform demand, and complete information in Section 4 allow for closed-form equilibrium characterizations. In many real-world competitive newsvendor environments, however, firms face heterogeneous cost structures and incomplete information about competitors. We therefore extend our analysis to a Bayesian version of the game, denoted by $\hat{\mathbf{G}}$, in which each player i privately observes their inventory cost c_i .

5.1. Bayesian profits and marginal incentives

Conditional on a realized cost type c_i and given inventory actions $\mathbf{y} = (y_i, y_j)$, player i 's expected profit is denoted by $\hat{J}_i(\mathbf{y}, c_i)$ taken from Equation (6). This profit function is obtained by taking expectations of the realized profit in (1) over stochastic demand and spillover effects, conditional on c_i . For equilibrium analysis and learning dynamics, it is convenient to work with the marginal profit function $\hat{I}_i(\mathbf{y}, c_i)$ defined in (8), which captures the local incentive of player i to increase inventory at a given cost realization.

Definition 5.1 (Bayes-Nash Equilibrium). A strategy profile $y^* = (y_1^*(\cdot), y_2^*(\cdot))$ is a Bayes-Nash equilibrium if, for

each player i and every realized cost type c_i , the action $y_i^*(c_i)$ maximizes expected profit $\hat{J}_i(\cdot, c_i)$ given the opponent's strategy and beliefs over the opponent's type.

Güler et al. (2018) derive a closed-form Bayes–Nash equilibrium for a model with *aggregate industry demand* and rationing, following the demand structure of Lippman & McCardle (1997). By contrast, for competitive newsvendor models with *firm-specific baseline demands* and stockout spillovers, as studied by Parlar (1988) and adopted in this paper, no closed-form Bayes–Nash equilibrium characterizations are available, even for simple demand and type distributions. Accordingly, in our numerical experiments we evaluate learned strategy profiles using equilibrium verification rather than analytical equilibrium predictions.

6. Strict Monotonicity

Equilibrium problems as in our setting can be written as a variational inequality (VI). We adopt the VI perspective because it allows us to connect equilibrium uniqueness and learning dynamics in a single operator-theoretic framework. The VI problem aims to find $y^* \in K$ such that

$$\langle F(y^*), y - y^* \rangle \geq 0 \quad \forall y \in K,$$

where K is a closed convex set and $F : K \rightarrow \mathbb{R}^d$. In games where agents *maximize* payoffs, a natural choice for F is

$$F(y) := -\nabla J(y),$$

where J denotes the vector of players' payoff functions, i.e., F is the game gradient. Under this convention, a standard structural assumption is *monotonicity*:

$$\langle F(y) - F(y'), y - y' \rangle \geq 0 \quad \forall y, y' \in K, \quad (11)$$

$$\Leftrightarrow (\nabla J(y) - \nabla J(y'))^\top (y - y') \leq 0. \quad (12)$$

Strict monotonicity strengthens this to a strict inequality for $y \neq y'$ and implies that a VI solution, if it exists, is *unique* (Rosen, 1965; Ui, 2016).

Monotonicity guarantees global asymptotic stability of the continuous-time payoff-gradient dynamics associated with the game (Mertikopoulos & Zhou, 2019). A variety of convergence results exist also for discrete-time dynamics. For monotone variational inequalities (hence also strictly monotone ones), globally convergent first-order methods include the proximal point algorithm (Rockafellar, 1976), Korpelevich's extragradient method (Korpelevich, 1977), Nemirovski's Mirror-Prox (Nemirovski, 2004), Tseng's forward–backward–forward splitting (Tseng, 2000), and the projected reflected gradient method (Malitsky, 2015). These methods are designed for monotone operators (including non-potential games; for our model see Appendix A) and achieve sublinear $O(1/k)$ rates under Lipschitz continuity (Nemirovski, 2004; Facchinei & Pang, 2003).

In our competitive newsvendor model, the game gradient can be written as $F(y) = -\mathbf{I}(y, \mathbf{c})$, where $\mathbf{I}(y, \mathbf{c})$ denotes the vector of marginal profit functions. Note that each component I_i is continuous. The shortage and salvage terms involve integrals of continuous functions, and the spillover threshold $B_i(y_i, x_i)$ depends linearly on y_i . Thus, each player's expected payoff is continuously differentiable in own inventory, and $\mathbf{J}$ is continuously differentiable on the interior of the action space. This regularity allows us to establish the following structural result. Even though in the previous results we kept parameter a general, we now focus on the most competitive case $a = 1$. This is the most demanding setting in which to study whether the game admits a monotone structure. Furthermore, it keeps the spillover terms tractable and the proofs transparent.

Theorem 6.1. *[Strict monotonicity of the competitive newsvendor game] Consider the two-player competitive newsvendor game with complete information and cost vector $\mathbf{c} = (c_1, c_2) \in \mathcal{C} \subset \mathbb{R}_+^2$. Assume that demand is uniformly distributed on $[0, 1]$, inventory levels are normalized such that the action set satisfies $K \subseteq [0, 1]^2$, and arbitrary parameters $(s, p, q) \in \mathbb{R}_+^3$ satisfy $s > c_i > q$ for $i = 1, 2$. Let*

$$\mathbf{I}(y, \mathbf{c}) := (I_1(y, c_1), I_2(y, c_2))$$

denote the marginal profit operator. Then, for every $\mathbf{c} \in \mathcal{C}$, the complete-information competitive newsvendor game is strictly monotone on K , i.e.,

$$\langle \mathbf{I}(y, \mathbf{c}) - \mathbf{I}(y', \mathbf{c}), y - y' \rangle < 0$$

for all (y, y') in K with $y \neq y'$.

The strict monotonicity proof is algebraic but conceptually simple: in every demand regime, cross-effects enter quadratically and are dominated by own-action curvature; the detailed proof can be found in Appendix C. To the best of our knowledge, this is the first strict monotonicity result for a competitive inventory game with demand spillovers. Let us now look at definitions of strict monotonicity for the Bayesian game. Here, $\hat{\mathbf{I}}(\cdot, \mathbf{c})$ is interpreted pointwise in types, i.e., as the collection of marginal profit operators evaluated at each realized cost vector $\mathbf{c}$.

Definition 6.2 (Ex-post strict monotonicity (Ui, 2016)). The competitive newsvendor game satisfies *ex-post strict monotonicity* if for almost every cost realization $\mathbf{c}$, the marginal profit operator $\hat{\mathbf{I}}(\cdot, \mathbf{c})$ satisfies

$$\langle \hat{\mathbf{I}}(y, \mathbf{c}) - \hat{\mathbf{I}}(y', \mathbf{c}), y - y' \rangle < 0 \quad \text{for all } y \neq y' \text{ in } K.$$

In words, a Bayesian game is ex-post strictly monotone in the sense of Ui (2016) if the payoff gradient is strictly monotone for almost every realization of the state. Theorem 6.1 shows that strict monotonicity holds for every $\mathbf{c}$, such that

the game is ex-post strictly monotone in the sense of [Ui \(2016\)](#).

For the convergence of extra-gradient or Mirror-Prox algorithms, an *ex-ante* monotonicity condition is sufficient. This is because learning algorithms do not observe a fixed realization of $\mathbf{c}$, but operate on expected or sampled gradients, and therefore interact with the expected operator $\bar{\mathbf{I}}$ (Equation (13)). Ex-post strict monotonicity in the sense of [Ui \(2016\)](#) is sufficient for equilibrium uniqueness but is stronger than needed for convergence of learning dynamics.

Definition 6.3 (Ex-ante strict monotonicity). The Bayesian competitive newsvendor game satisfies *ex-ante strict monotonicity* if the expected marginal profit operator

$$\bar{\mathbf{I}}(\mathbf{y}) := \mathbb{E}_{\mathbf{c}} [\hat{\mathbf{I}}(\mathbf{y}, \mathbf{c})] \quad (13)$$

is strictly monotone, i.e.,

$$\langle \bar{\mathbf{I}}(\mathbf{y}) - \bar{\mathbf{I}}(\mathbf{y}'), \mathbf{y} - \mathbf{y}' \rangle < 0 \quad \text{for all } \mathbf{y} \neq \mathbf{y}'. \quad (14)$$

Note that if strict monotonicity holds ex-post, then ex-ante strict monotonicity also holds. This can be seen with

$$\begin{aligned} & \langle \mathbb{E}_{\mathbf{c}} [\hat{\mathbf{I}}(\mathbf{y}, \mathbf{c})] - \mathbb{E}_{\mathbf{c}} [\hat{\mathbf{I}}(\mathbf{y}', \mathbf{c})], \mathbf{y} - \mathbf{y}' \rangle < 0 \\ \Leftrightarrow & \quad \mathbb{E}_{\mathbf{c}} [\langle \hat{\mathbf{I}}(\mathbf{y}, \mathbf{c}) - \hat{\mathbf{I}}(\mathbf{y}', \mathbf{c}), \mathbf{y} - \mathbf{y}' \rangle] < 0 \end{aligned}$$

The implications follows from the linearity of the expectation and the scalar product.

Strict monotonicity implies that deviations from equilibrium are globally disciplined and rules out cycling behavior. As a result, the equilibrium is not only unique but also stable under a broad class of adaptive dynamics. While policy-gradient methods such as PPO are not designed to solve variational inequalities, their updates can be interpreted as zeroth-order approximated gradient steps on the underlying game operator. In a strictly monotone environment, such updates are naturally aligned with equilibrium incentives, providing an explanation for the stable convergence observed in our experiments

7. Numerical Results

Our theoretical analysis establishes strict monotonicity of the competitive newsvendor game, implying equilibrium uniqueness and ruling out cyclic or unstable dynamics. However, formal convergence guarantees apply only to specific classes of variational-inequality solvers, such as extragradient or Mirror-Prox methods, which explicitly exploit monotonicity. Modern DRL algorithms, by contrast, are not designed to solve variational inequalities and lack convergence guarantees even in simple general-sum games. Yet, such algorithms are widely used to automate inventory management decisions in firms.

The goal of this section is therefore not to establish new algorithmic guarantees, but to empirically test the following hypothesis: *whether independent, DRL-based learning agents converge to equilibrium in strictly monotone continuous-action games*. To this end, we study self-play learning dynamics using DRL in both a complete-information setting with fixed costs and a Bayesian setting where costs of competitors vary. Additionally, we compare the uniform demand distribution to a Gaussian demand setting to assess robustness across different demand environments.

7.1. DRL Implementation and Verification

We employ Proximal Policy Optimization (PPO) ([Schulman et al., 2015](#)) as a representative policy-gradient method for continuous action spaces. PPO is not designed to exploit monotonicity or variational-inequality structure explicitly; rather, it implements a stabilized, clipped policy-gradient update with stochastic sampling and projection onto a constrained policy class. As such, PPO serves as a natural stress test for whether DRL-based learners converge in competitive newsvendor games.

The goal of this paper is not to show that PPO is the best algorithm for this problem, but to test whether a widely used DRL method can learn equilibrium strategies in a strictly monotone game. We also report results for REINFORCE and TD3 in Appendix E to show that convergence is not specific to PPO.

All agents are trained independently using self-play. Hyperparameters are held fixed across all experiments and are reported in Table 3 in Appendix D. Each experiment is repeated with 10 random seeds to assess robustness. Across all scenarios, the standard deviation of both learned actions and verified best responses remains below 2% of the mean value, indicating highly stable convergence independent of random initialization.

To evaluate convergence, we do not rely on reward stabilization alone. Instead, we verify learned policies using the equilibrium verifier of [Pieroth et al. \(2025\)](#), which computes the maximal unilateral deviation gain for each agent via Monte Carlo simulation of demand and cost draws. This equilibrium-based evaluation is particularly important in general-sum games, where stable rewards do not imply proximity to equilibrium. Across all scenarios considered, learned policies satisfy increasingly tight equilibrium conditions over training and converge to the unique (Bayes-)Nash equilibrium predicted by theory.

7.2. Complete-Information Game

We begin with the symmetric complete-information game $\mathbf{G}^*$ introduced in Section 4. This setting provides a controlled baseline in which equilibrium predictions are avail-

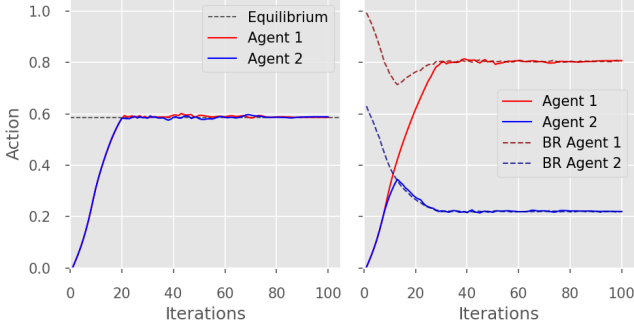


Figure 1. Evolution of agent actions during PPO training in symmetric (left) and asymmetric (right) complete-information games, benchmarked against Nash equilibria / estimated best responses.

able in closed form and learning dynamics can be directly compared to theoretical benchmarks. Agents observe a selling price of $s = 1$, a cost of $c = 0.5$, and zero penalty and salvage values ($p = q = 0$). Demands are independently drawn from a uniform distribution $x_i \sim U[0, 1]$.

Figure 1 illustrates the evolution of agents’ actions under independent PPO learning. Within approximately 20 training iterations—each consisting of 20,000 environment steps—both agents’ policies rapidly approach the unique symmetric Nash equilibrium $y_i \approx 0.5858$ predicted by Theorem 4.1. Subsequent updates produce only small fluctuations around this point, indicating stable convergence rather than cycling or drift. After 100 iterations, the learned actions remain tightly concentrated around the equilibrium. The remaining deviations reflect approximation error due to finite samples and neural-network policy classes rather than systematic bias. For visual clarity, the figure displays mean trajectories only; standard deviation bands across seeds are omitted as they remain below 2% throughout training and would be barely visible at the scale shown.

We next consider an asymmetric cost configuration with $c_1 = 0.5$ and $c_2 = 0.8$. Since no closed-form solution is available in this case, we compute the benchmark via an exhaustive grid-based best-response (BR) search (see Appendix D), yielding $(y_1^*, y_2^*) \approx (0.8046, 0.2191)$.

The right panel of Figure 1 shows the corresponding learning dynamics. Early in training, both agents increase their actions as gradient estimates reflect high expected surplus demand. As learning progresses, the effective marginal incentive for agent 2 weakens due to its higher cost, leading to a gradual downward adjustment of its policy. After approximately 40 iterations, both agents stabilize near a stable point, with learned actions closely matching the computed estimated best responses. Table 1 in Appendix D summarizes the numerical outcomes after 100 iterations; relative errors remain below 0.4%.

Finally, we examine how penalty and salvage parameters af-

fect equilibrium learning. Raising either the penalty p or the salvage value q to 0.2 increases joint stocking beyond the expected total demand of 1.0. Introducing a penalty p primarily strengthens the high-cost player’s incentive to stock more, which reduces the expected surplus demand faced by the other player and allows him to lower his stocking level. Conversely, introducing a salvage value q largely eliminates the low-cost player’s downside risk from unsold inventory, leading him to increase stocking substantially. Increasing both parameters raises stocking for both players, with the strongest increase occurring for the high-cost player.

7.3. Bayesian Game

We next study learning dynamics in the Bayesian version of the competitive newsvendor game described in Section 5. In this setting, agents face incomplete information and must learn *type-contingent policies* rather than single inventory levels. Each agent observes a selling price of $s = 1$ and zero penalty and salvage values ($p = q = 0$), while its cost parameter is drawn independently from a commonly known distribution at the beginning of each episode.

We examine two Bayesian scenarios. One (symmetric) case, where both agents draw costs from $L_i = \{40\% : 0.3, 60\% : 0.7\}$ and one (asymmetric) case where agent 1 draws from the same distribution and agent 2 from $L_2 = \{50\% : 0.2, 50\% : 0.8\}$. In both settings, agents must learn type-contingent policies mapping private costs to inventory actions. In both scenarios, we use the standard uniform demand specification $U(0, 1)$ and additionally consider with the same set of parameters a Gaussian demand distribution $\mathcal{N}(0.5, 0.15)$. This allows us to compare performance across different demand distributions.

Despite the increased complexity of learning over type spaces, independent PPO agents converge reliably to Bayes–Nash equilibrium strategies in both scenarios. As summarized in Table 2 in Appendix D, the learned type-contingent policies closely match computed best responses after 400 training iterations in both demand settings, with relative errors below 1% in all cases. No oscillatory or divergent behavior is observed across random seeds.

From a learning perspective, these results indicate that strict monotonicity extends its stabilizing effect beyond complete-information Nash equilibria to settings where agents must learn equilibrium *functions* over private information. Even though PPO does not exploit the Hilbert-space structure of Bayesian games explicitly, the induced learning dynamics remain stable and converge to the unique BNE predicted by theory.

8. Conclusion

This paper studies when and why independent learning agents converge in continuous-action general-sum games. Focusing on the competitive newsvendor game with uniformly distributed demand as a canonical testbed, we identify a strong structural property—*strict monotonicity* of the payoff gradient—that fundamentally shapes learning dynamics. Despite strategic substitutability and the absence of potential structure, strict monotonicity rules out cyclic or chaotic behavior and guarantees equilibrium uniqueness. From a learning perspective, it places the game in a class where first-order methods for variational inequalities admit global convergence guarantees, a property that is rarely available in competitive environments.

Our empirical results complement this structural analysis. Using Proximal Policy Optimization in self-play, we find that independent deep reinforcement learning agents reliably converge to Nash and Bayes–Nash equilibria across symmetric and asymmetric settings, including environments with private information. Although PPO is not designed to exploit monotonicity explicitly and lacks theoretical convergence guarantees in general-sum games, its learning dynamics remain stable in this setting and converge to the equilibria predicted by theory. Equilibrium verification confirms that convergence reflects genuine strategic stability rather than reward plateaus.

Taken together, our results help explain why learning succeeds in competitive newsvendor games with uniform demand while failing in other competitive models, such as oligopoly and auction environments, where monotonicity is violated and learning dynamics are known to cycle or diverge. More broadly, this work highlights monotonicity as a key structural condition linking equilibrium theory and learning dynamics in multi-agent systems. We view these findings as a step toward a more systematic understanding of which classes of games are amenable to reliable learning with modern gradient-based and deep reinforcement learning methods.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Non-Potential Game

Lemma A.1 (The complete-information game is non-potential). *Assume the standard pricing condition $s > q$. Then the complete-information competitive newsvendor game is not an exact potential game on $K_1 \times K_2$.*

Proof. The expected payoff functions $J_i(\mathbf{y}, c_i)$ are C^2 in (y_1, y_2) on the interior of the compact action set. By Theorem 4.5 in [Monderer & Shapley \(1996\)](#), a necessary and sufficient condition for exact potentiality in this setting is

$$\frac{\partial^2 J_1}{\partial y_1 \partial y_2}(\mathbf{y}) = \frac{\partial^2 J_2}{\partial y_2 \partial y_1}(\mathbf{y}) \quad \text{for all } \mathbf{y} \in K_1 \times K_2. \quad (15)$$

For the competitive newsvendor payoff structure used in the paper, these cross-partials are

$$\frac{\partial^2 J_1}{\partial y_1 \partial y_2}(\mathbf{y}) = -(s - q)y_1, \quad \frac{\partial^2 J_2}{\partial y_2 \partial y_1}(\mathbf{y}) = -(s - q)y_2. \quad (16)$$

Since $s > q$, equality would require $y_1 = y_2$ at every profile, which is false on the two-dimensional set $K_1 \times K_2$. Hence the integrability condition fails globally, and the game is not an exact potential game. $\square$

B. Symmetric Competitive Newsvendor Game

Theorem 4.1. *In the simplified symmetric competitive newsvendor game $\mathbf{G}^*$ with $p = 0$, $q = 0$, and $a = 1$, where demand is uniformly distributed on $[0, 1]$, the unique symmetric Nash equilibrium inventory level for both players is given by*

$$y_i = \begin{cases} 2 - \sqrt{1 + 2\frac{c}{s}}, & \text{if } \frac{c}{s} \leq \frac{5}{8}, \\ \sqrt{\frac{2}{3}} \left(1 - \frac{c}{s}\right), & \text{if } \frac{c}{s} > \frac{5}{8}. \end{cases} \quad (9)$$

Proof. To characterize equilibrium in this symmetric benchmark, we start from each firm's first-order optimality condition. Building on Equation (3) and the derivation in [Parlar \(1988\)](#) (p. 402), the Nash equilibrium is pinned down by

$$I_i(\mathbf{y}, c) = \frac{\partial J_i}{\partial y_i}(\mathbf{y}, c) = 0,$$

where I_i represents player i 's marginal profit with respect to its inventory choice.

For a given inventory profile $\mathbf{y} = (y_1, y_2)$, this marginal profit takes the form as in Equation (4):

$$\begin{aligned} I_i(\mathbf{y}, c) = & (s + p) \int_{y_i}^{\infty} f(x_i) dx_i + q \int_0^{y_i} f(x_i) dx_i \\ & + (s - q) \int_0^{y_i} \int_{B(x_i)}^{\infty} g(x_j) f(x_i) dx_j dx_i - c, \end{aligned}$$

where $B(x_i) = \frac{y_i - x_i}{a} + y_j$ denotes the spillover threshold that determines when rival demand is transferred.

We now specialize to uniform demand $x_i \sim \mathcal{U}[0, 1]$ and the simplified parameterization $p = q = 0$, $a = 1$, in which case the marginal profit simplifies to

$$I_i(\mathbf{y}, c) = s(1 - y_i) - c + s \int_0^{y_i} (1 - y_i + x_i - y_j)^+ dx_i.$$

Because the game is symmetric and the equilibrium is unique, it suffices to focus on symmetric profiles; hence we restrict attention to $y_1 = y_2 = y$. The equilibrium characterization then depends on whether total inventory $2y$ is above or below the expected total demand 1.

High-inventory regime ($y \geq \frac{1}{2}$). When $y \geq \frac{1}{2}$, the relevant integral truncates, and the condition $I_i(y, c) = 0$ becomes

$$I_i(y) = s(1 - y) - c + s \int_{2y-1}^y (1 - 2y + x) dx = s(1 - y) - c + s \int_0^{1-y} u du = s(1 - y) - c + \frac{s}{2}(1 - y)^2 = 0.$$

Here we substitute $u := 1 - 2y + x$, so $u = 0$ at $x = 2y - 1$ and $u = 1 - y$ at $x = y$. This reduces to a quadratic in $z := 1 - y$,

$$\begin{aligned} 0 &= \frac{s}{2}z^2 + sz - c \\ \iff z &= \frac{-s \pm \sqrt{s^2 + 2sc}}{s} = -1 \pm \sqrt{1 + 2\frac{c}{s}}. \end{aligned}$$

Nonnegativity of inventory implies $z = -1 + \sqrt{1 + 2\frac{c}{s}}$, which gives

$$y = 2 - \sqrt{1 + 2\frac{c}{s}}.$$

This candidate is valid precisely when $y \geq \frac{1}{2}$, i.e., when $\frac{c}{s} \leq \frac{5}{8}$.

Low-inventory regime ($y < \frac{1}{2}$). If $y < \frac{1}{2}$, the lower integration bound is zero and the first-order condition simplifies to

$$I_i(y) = s(1 - y) - c + s \int_0^y (1 - 2y + x) dx = 0$$

Evaluating the integral gives

$$\int_0^y (1 - 2y + x) dx = \left[(1 - 2y)x + \frac{x^2}{2} \right]_0^y = y - 2y^2 + \frac{y^2}{2} = y - \frac{3}{2}y^2,$$

so

$$I_i(y) = s(1 - y) - c + s \left(y - \frac{3}{2}y^2 \right) = s - c - \frac{3s}{2}y^2 = 0.$$

Solving for y yields $y = \sqrt{\frac{2(s-c)}{3s}}$, which applies when $\frac{c}{s} > \frac{5}{8}$. □

Theorem 4.2. Consider the symmetric competitive newsvendor game $\mathbf{G}$ with arbitrary parameters $(p, q, s, c, a) \in \mathbb{R}_+^5$ satisfying $s > q$ and $a > 0$, where demand is uniformly distributed on $[0, 1]$. The unique symmetric Nash equilibrium inventory level satisfies the following. If

$$s + p - c \leq \frac{ap}{a+1} + \frac{a(2a+1)}{2(a+1)^2}(s-q), \quad (10)$$

then

$$y_i = \frac{-ap + \sqrt{(ap)^2 + 2a(s+p-c)(s-q)(2a+1)}}{(s-q)(2a+1)}.$$

Otherwise,

$$y_i = 1 + \frac{1}{a} + \frac{p}{a(s-q)} - \frac{\sqrt{(p+s-q)^2 + 2a(s-q)(c-q)}}{a(s-q)}.$$

Proof. We now turn to the general parameter setting. The proof follows the same structure as the symmetric benchmark, but the marginal profit expression is richer and must be handled piecewise.

As in the previous proof, we characterize the unique equilibrium following [Parlar \(1988\)](#). The first-order condition can be written as:

$$\begin{aligned} I_i(\mathbf{y}) &= \frac{\partial J_i}{\partial y_i}(y_1, y_2) \\ &= (s+p) \int_{y_i}^{\infty} f(x_i) dx_i + q \int_0^{y_i} f(x_i) dx_i + (s-q) \int_0^{y_i} \int_{B(x_i)}^{\infty} g(x_j) f(x_i) dx_j dx_i - c \\ &= (s+p)(1-y_i) + qy_i - c + (s-q) \begin{cases} y_i \left(1 - y_j - \frac{y_i}{2a}\right), & \text{if } \frac{1}{a}y_i + y_j \leq 1, \\ \frac{a}{2}(1-y_j)^2, & \text{otherwise.} \end{cases} \end{aligned} \quad (17)$$

$$\begin{aligned} &= (s+p)(1-y) + qy - c + (s-q) \begin{cases} y \left(1 - y \left(1 + \frac{1}{2a}\right)\right), & \text{if } y \left(1 + \frac{1}{a}\right) \leq 1, \\ \frac{a}{2}(1-y)^2, & \text{otherwise.} \end{cases} \\ &= 0 \end{aligned} \quad (18)$$

with $B(x_1) = \frac{y_1 - x_1}{a} + y_2$. We again restrict attention to symmetric equilibria and set $y := y_1 = y_2$.

Case $y(1 + \frac{1}{a}) \leq 1$ In the regime where the spillover threshold remains below the unit demand bound, Equation (18) reduces to a quadratic in y :

$$\begin{aligned} 0 &= (s+p)(1-y) + qy - c + (s-q)y \left(1 - y \left(1 + \frac{1}{2a}\right)\right) \\ &= (s+p-c) - py - (s-q)\left(1 + \frac{1}{2a}\right)y^2 \\ &= \frac{(s-q)(2a+1)}{2a}y^2 + py - (s+p-c) \end{aligned}$$

which yields two candidate solutions,

$$y = \frac{-p \pm \sqrt{T}}{2(s-q) \left(1 + \frac{1}{2a}\right)},$$

where $T = p^2 + 4(s+p-c)(s-q) \left(1 + \frac{1}{2a}\right)$. Since $T \geq p^2$, the nonnegative equilibrium is

$$\begin{aligned} y &= \frac{-p + \sqrt{T}}{2(s-q) \left(1 + \frac{1}{2a}\right)} \\ &= \frac{-ap + \sqrt{(ap)^2 + 2a(s+p-c)(s-q)(2a+1)}}{(s-q)(2a+1)}. \end{aligned}$$

Enforcing the regime condition yields a parameter restriction:

$$\begin{aligned} &y \left(1 + \frac{1}{a}\right) \leq 1 \\ \iff &\frac{-p + \sqrt{T}}{2(s-q) \left(1 + \frac{1}{2a}\right)} \leq \frac{a}{a+1} \\ \iff &\sqrt{T} \leq p + \frac{2a}{a+1}(s-q) \left(1 + \frac{1}{2a}\right) \\ \iff &T \leq \left(p + \frac{2a+1}{a+1}(s-q)\right)^2 \\ \iff &p^2 + 4(s+p-c)(s-q) \left(1 + \frac{1}{2a}\right) \leq p^2 + \frac{2p(2a+1)}{a+1}(s-q) + \left(\frac{2a+1}{a+1}\right)^2 (s-q)^2 \\ \iff &4(s+p-c) \left(\frac{2a+1}{2a}\right) \leq \frac{2p(2a+1)}{a+1} + \left(\frac{2a+1}{a+1}\right)^2 (s-q) \\ \iff &s+p-c \leq \frac{ap}{a+1} + \frac{a(2a+1)}{2(a+1)^2}(s-q). \end{aligned}$$

This gives the desired condition.

Case $y(1 + \frac{1}{a}) > 1$: In the complementary regime, the first-order condition from Equation (18) simplifies to

$$\begin{aligned} &(s+p)(1-y) + qy - c + (s-q) \frac{a(1-y)^2}{2} = 0 \\ \iff &(s+p)y' + q - qy' - c + a(s-q) \frac{y'^2}{2} = 0 \\ \iff &\frac{a(s-q)}{2} y'^2 + (s+p-q)y' + q - c = 0. \end{aligned}$$

with the substitution $y' := 1 - y$.

Solving this quadratic in y' , we obtain

$$y' = \frac{-(s+p-q) \pm \sqrt{(s+p-q)^2 - 2a(s-q)(q-c)}}{a(s-q)}$$

and for y ,

$$\begin{aligned}
 y &= 1 - \frac{-(s+p-q) + \sqrt{(s+p-q)^2 + 2a(s-q)(c-q)}}{a(s-q)} \\
 &= 1 + \frac{s+p-q}{a(s-q)} - \frac{\sqrt{(s+p-q)^2 + 2a(s-q)(c-q)}}{a(s-q)} \\
 &= 1 + \frac{(s-q)+p}{a(s-q)} - \frac{\sqrt{(p+s-q)^2 + 2a(s-q)(c-q)}}{a(s-q)} \\
 &= 1 + \frac{1}{a} + \frac{p}{a(s-q)} - \frac{\sqrt{(p+s-q)^2 + 2a(s-q)(c-q)}}{a(s-q)},
 \end{aligned}$$

which gives us the second candidate solution. □

C. Strict Monotonicity

Theorem 6.1. *[Strict monotonicity of the competitive newsvendor game] Consider the two-player competitive newsvendor game with complete information and cost vector $\mathbf{c} = (c_1, c_2) \in \mathcal{C} \subset \mathbb{R}_+^2$. Assume that demand is uniformly distributed on $[0, 1]$, inventory levels are normalized such that the action set satisfies $K \subseteq [0, 1]^2$, and arbitrary parameters $(s, p, q) \in \mathbb{R}_+^3$ satisfy $s > c_i > q$ for $i = 1, 2$. Let*

$$\mathbf{I}(\mathbf{y}, \mathbf{c}) := (I_1(\mathbf{y}, c_1), I_2(\mathbf{y}, c_2))$$

denote the marginal profit operator. Then, for every $\mathbf{c} \in \mathcal{C}$, the complete-information competitive newsvendor game is strictly monotone on K , i.e.,

$$\langle \mathbf{I}(\mathbf{y}, \mathbf{c}) - \mathbf{I}(\mathbf{y}', \mathbf{c}), \mathbf{y} - \mathbf{y}' \rangle < 0$$

for all $(\mathbf{y}, \mathbf{y}')$ in K with $\mathbf{y} \neq \mathbf{y}'$.

Proof. Consider two distinct strategy profiles $\mathbf{y} = (y_1, y_2)$ and $\mathbf{y}' = (y'_1, y'_2)$ in the strategy space K . To establish strict monotonicity, we must verify that the variational inequality holds strictly.

Define $\Delta_i := y_i - y'_i$ as the inventory difference for player i , and let $T_i := I_i(\mathbf{y}, c_i) - I_i(\mathbf{y}', c_i)$ denote the corresponding marginal profit difference.

The monotonicity condition is encoded in the function:

$$F(\mathbf{y}, \mathbf{y}', \mathbf{c}) := \left\langle \begin{pmatrix} I_1(\mathbf{y}, c_1) \\ I_2(\mathbf{y}, c_2) \end{pmatrix} - \begin{pmatrix} I_1(\mathbf{y}', c_1) \\ I_2(\mathbf{y}', c_2) \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} - \begin{pmatrix} y'_1 \\ y'_2 \end{pmatrix} \right\rangle = T_1 \Delta_1 + T_2 \Delta_2,$$

which must be shown strictly negative whenever $\mathbf{y} \neq \mathbf{y}'$.

Recall from Equation 17 that player i 's marginal profit takes the form:

$$I_i(\mathbf{y}, c_i) = (s + p)(1 - y_i) + qy_i - c_i + (s - q) \begin{cases} y_i(1 - y_j - \frac{1}{2}y_i), & \text{if } y_1 + y_2 \leq 1, \\ \frac{1}{2}(1 - y_j)^2, & \text{otherwise.} \end{cases}$$

The piecewise structure of this marginal profit induces a natural partition of the strategy space. Depending on whether the aggregate inventories $y_1 + y_2$ and $y'_1 + y'_2$ lie above or below the demand threshold of 1, we distinguish four cases:

1. $y_1 + y_2 \leq 1 \quad \wedge \quad y'_1 + y'_2 \leq 1$
2. $y_1 + y_2 > 1 \quad \wedge \quad y'_1 + y'_2 > 1$
3. $y_1 + y_2 \leq 1 \quad \wedge \quad y'_1 + y'_2 > 1$
4. $y_1 + y_2 > 1 \quad \wedge \quad y'_1 + y'_2 \leq 1$

We will establish $F(\mathbf{y}, \mathbf{y}', \mathbf{c}) < 0$ separately in each region, thereby proving strict monotonicity across the entire strategy space.

Case 1: Low inventory regime ($y_1 + y_2 \leq 1$ and $y'_1 + y'_2 \leq 1$)

Both strategy profiles lie below the demand threshold. Player i 's marginal profit is

$$I_i(\mathbf{y}, c_i) = (s + p)(1 - y_i) + qy_i - c_i + (s - q)y_i\left(1 - y_j - \frac{1}{2}y_i\right).$$

Taking differences yields

$$\begin{aligned} T_i &:= I_i(\mathbf{y}, c_i) - I_i(\mathbf{y}', c_i) \\ &= (q - s - p)\Delta_i + (s - q)\left[y_i\left(1 - y_j - \frac{1}{2}y_i\right) - y'_i\left(1 - y'_j - \frac{1}{2}y'_i\right)\right]. \end{aligned}$$

Multiplying by Δ_i and summing over both players, the monotonicity function becomes

$$\begin{aligned} F(\mathbf{y}, \mathbf{y}', \mathbf{c}) &= \sum_{i=1}^2 T_i \Delta_i \\ &= (q - s - p)(\Delta_1^2 + \Delta_2^2) + (s - q) \sum_{i=1}^2 \left[y_i \left(1 - y_j - \frac{1}{2} y_i \right) - y'_i \left(1 - y'_j - \frac{1}{2} y'_i \right) \right] \Delta_i \\ &\leq -(s - q)(\Delta_1^2 + \Delta_2^2) + (s - q) \sum_{i=1}^2 (\phi_{1-y_j}(y_i) - \phi_{1-y'_j}(y'_i)) \Delta_i \end{aligned}$$

where we introduce the quadratic function $\phi_r(x) := rx - \frac{1}{2}x^2$ with parameter r .

Since $p \geq 0$ and $s > q$, the coefficient satisfies $q - s - p \leq -(s - q)$, allowing us to bound

$$\begin{aligned} F(\mathbf{y}, \mathbf{y}', \mathbf{c}) &= (q - s - p)(\Delta_1^2 + \Delta_2^2) + (s - q) \sum_{i=1}^2 (\phi_{1-y_j}(y_i) - \phi_{1-y'_j}(y'_i)) \Delta_i \\ &\leq (s - q) \left[-(\Delta_1^2 + \Delta_2^2) + \sum_{i=1}^2 (\phi_{1-y_j}(y_i) - \phi_{1-y'_j}(y'_i)) \Delta_i \right]. \end{aligned}$$

Because $s - q > 0$, it suffices to verify the strict inequality

$$\sum_{i=1}^2 (\phi_{1-y_j}(y_i) - \phi_{1-y'_j}(y'_i)) \Delta_i < \Delta_1^2 + \Delta_2^2. \quad (19)$$

This inequality is verified by direct algebraic expansion. Lemma C.1 establishes the identity

$$\sum_{i=1}^2 (\phi_{1-y_j}(y_i) - \phi_{1-y'_j}(y'_i)) \Delta_i = \Delta_1^2 + \Delta_2^2 - \frac{y_1 + y_2 + y'_1 + y'_2}{4} ((\Delta_1 + \Delta_2)^2 + \Delta_1^2 + \Delta_2^2).$$

Since $\mathbf{y} \neq \mathbf{y}'$, at least one difference Δ_i is nonzero, and at least one inventory level among y_1, y_2, y'_1, y'_2 is positive. Consequently, both the coefficient $\frac{1}{4}(y_1 + y_2 + y'_1 + y'_2)$ and the quadratic form $(\Delta_1 + \Delta_2)^2 + \Delta_1^2 + \Delta_2^2$ are strictly positive. This establishes the strict inequality (19), yielding $F(\mathbf{y}, \mathbf{y}', \mathbf{c}) < 0$ and completing Case 1.

Case 2: High inventory regime ($y_1 + y_2 > 1$ and $y'_1 + y'_2 > 1$)

Both profiles now exceed the demand threshold. The marginal profit simplifies to

$$I_i(\mathbf{y}, c_i) = (s + p)(1 - y_i) + qy_i - c_i + (s - q)\frac{1}{2}(1 - y_j)^2.$$

The marginal profit difference becomes

$$\begin{aligned} T_i &= I_i(\mathbf{y}, c_i) - I_i(\mathbf{y}', c_i) = (q - s - p)(y_i - y'_i) + (s - q)\frac{1}{2}((1 - y_j)^2 - (1 - y'_j)^2), \\ T_i \Delta_i &= (q - s - p)\Delta_i^2 + (s - q)\frac{1}{2}((1 - y_j)^2 - (1 - y'_j)^2)\Delta_i. \end{aligned}$$

We simplify the quadratic term using the factorization

$$(1 - y_j)^2 - (1 - y'_j)^2 = ((1 - y_j) - (1 - y'_j))((1 - y_j) + (1 - y'_j)) = (y'_j - y_j)(2 - y_j - y'_j) = -\Delta_j(2 - y_j - y'_j).$$

Substituting and summing over both players yields

$$\begin{aligned} F(\mathbf{y}, \mathbf{y}', \mathbf{c}) &= (q - s - p)(\Delta_1^2 + \Delta_2^2) - (s - q)\frac{1}{2}\Delta_1\Delta_2(4 - y_1 - y_2 - y'_1 - y'_2), \\ &\leq -(s - q)(\Delta_1^2 + \Delta_2^2) - (s - q)\frac{1}{2}\Delta_1\Delta_2(4 - y_1 - y_2 - y'_1 - y'_2). \end{aligned}$$

Since $s - q > 0$, strict negativity holds if

$$\Delta_1^2 + \Delta_2^2 > -\frac{1}{2}\Delta_1\Delta_2(4 - y_1 - y_2 - y'_1 - y'_2). \quad (20)$$

Since $\mathbf{y} \neq \mathbf{y}'$, at least one Δ_i is nonzero. Moreover, the aggregate inventory constraint implies $4 - y_1 - y_2 - y'_1 - y'_2 \in (0, 2)$. We distinguish two subcases based on the sign of $\Delta_1\Delta_2$:

If $\Delta_1\Delta_2 \geq 0$, the right side is nonpositive while the left side is strictly positive, so inequality (20) holds trivially.

If $\Delta_1\Delta_2 < 0$, the AM-GM inequality $\Delta_1^2 + \Delta_2^2 \geq 2|\Delta_1\Delta_2|$ gives

$$\Delta_1^2 + \Delta_2^2 \geq 2|\Delta_1\Delta_2| = -\Delta_1\Delta_2 \frac{4}{2} > -\frac{1}{2}\Delta_1\Delta_2(4 - y_1 - y_2 - y'_1 - y'_2).$$

Thus inequality (20) holds in both subcases, establishing $F(\mathbf{y}, \mathbf{y}', \mathbf{c}) < 0$ and completing Case 2.

Case 3: Mixed inventory regimes ($y_1 + y_2 \leq 1$ and $y'_1 + y'_2 > 1$).

The profiles now straddle the demand threshold, inducing distinct marginal profit expressions:

$$\begin{aligned} I_i(\mathbf{y}, c_i) &= (s + p)(1 - y_i) + qy_i - c_i + (s - q)y_i\left(1 - y_j - \frac{y_i}{2}\right), \\ I_i(\mathbf{y}', c_i) &= (s + p)(1 - y'_i) + qy'_i - c_i + (s - q)\frac{1}{2}(1 - y'_j)^2. \end{aligned}$$

We introduce an auxiliary term to capture the asymmetry:

$$B_i := y_i\left(1 - y_j - \frac{y_i}{2}\right) - \frac{1}{2}(1 - y'_j)^2.$$

This allows us to express the marginal profit difference and monotonicity function as

$$\begin{aligned} T_i &= (q - s - p)\Delta_i + (s - q)B_i, \\ F(\mathbf{y}, \mathbf{y}', \mathbf{c}) &= (q - s - p)(\Delta_1^2 + \Delta_2^2) + (s - q)(B_1\Delta_1 + B_2\Delta_2) \\ &\leq -(s - q)(\Delta_1^2 + \Delta_2^2) + (s - q)\sum_{i=1}^2 B_i\Delta_i. \end{aligned}$$

As $s > q$, strict negativity follows if

$$\sum_{i=1}^2 B_i\Delta_i < \Delta_1^2 + \Delta_2^2. \quad (21)$$

Lemma C.2 provides the expansion

$$\sum_{i=1}^2 B_i\Delta_i = \Delta_1^2 + \Delta_2^2 - \frac{y_1 + y_2 + y'_1 + y'_2}{4}\left((\Delta_1 + \Delta_2)^2 + \Delta_1^2 + \Delta_2^2\right) - \frac{(y'_1 + y'_2 - 1)^2}{2}(\Delta_1 + \Delta_2).$$

Thus, inequality (21) reduces to verifying

$$\begin{aligned} &\frac{y_1 + y_2 + y'_1 + y'_2}{4}\left((\Delta_1 + \Delta_2)^2 + \Delta_1^2 + \Delta_2^2\right) + \frac{(y'_1 + y'_2 - 1)^2}{2}(\Delta_1 + \Delta_2) > 0 \\ \Leftrightarrow &\frac{y_1 + y_2 + y'_1 + y'_2}{4}\left((\Delta_1 + \Delta_2)^2 + \Delta_1^2 + \Delta_2^2\right) > -\frac{(y'_1 + y'_2 - 1)^2}{2}(\Delta_1 + \Delta_2) \end{aligned}$$

In this regime, $y'_1 + y'_2 > 1$ and $y_1 + y_2 \leq 1$ imply

$$\Delta_1 + \Delta_2 = (y_1 + y_2) - (y'_1 + y'_2) \leq 1 - (y'_1 + y'_2) < 0, \quad (22)$$

Since $\mathbf{y} \neq \mathbf{y}'$, we have $\Delta_1^2 + \Delta_2^2 > 0$. The desired inequality then simplifies to

$$\begin{aligned}
 & \frac{y_1 + y_2 + y'_1 + y'_2}{4} \left((\Delta_1 + \Delta_2)^2 + \Delta_1^2 + \Delta_2^2 \right) > -\frac{(y'_1 + y'_2 - 1)^2}{2} (\Delta_1 + \Delta_2) \\
 \Leftrightarrow & \frac{y_1 + y_2 + y'_1 + y'_2}{4} (\Delta_1 + \Delta_2)^2 + \frac{y_1 + y_2 + y'_1 + y'_2}{4} (\Delta_1^2 + \Delta_2^2) > -\frac{(y'_1 + y'_2 - 1)^2}{2} (\Delta_1 + \Delta_2) \\
 \Leftarrow & \frac{y_1 + y_2 + y'_1 + y'_2}{4} (-(\Delta_1 + \Delta_2))^2 \geq \frac{(y'_1 + y'_2 - 1)^2}{2} (-(\Delta_1 + \Delta_2)) \\
 \Leftrightarrow & \frac{y_1 + y_2 + y'_1 + y'_2}{4} (-(\Delta_1 + \Delta_2)) \geq \frac{(y'_1 + y'_2 - 1)^2}{2}
 \end{aligned}$$

Here we dropped the strictly positive term $(y_1 + y_2 + y'_1 + y'_2)(\Delta_1^2 + \Delta_2^2)/4$ to obtain a sufficient condition. Using $-(\Delta_1 + \Delta_2) \geq y'_1 + y'_2 - 1$ (from (22)) and $y_1 + y_2 + y'_1 + y'_2 \geq y'_1 + y'_2$, we verify

$$\begin{aligned}
 & \frac{y_1 + y_2 + y'_1 + y'_2}{4} (-(\Delta_1 + \Delta_2)) \geq \frac{(y'_1 + y'_2 - 1)^2}{2} \\
 \Leftarrow & \frac{y'_1 + y'_2}{4} (y'_1 + y'_2 - 1) \geq \frac{(y'_1 + y'_2 - 1)^2}{2} \\
 \Leftrightarrow & \frac{1}{2} (y'_1 + y'_2) \geq y'_1 + y'_2 - 1 \\
 \Leftrightarrow & y'_1 + y'_2 \leq 2
 \end{aligned}$$

This holds automatically since $y'_i \leq 1$ for each i .

Note that we strengthened the inequality by discarding the positive term

$$\frac{y_1 + y_2 + y'_1 + y'_2}{4} (\Delta_1^2 + \Delta_2^2),$$

which is strictly positive since $\mathbf{y} \neq \mathbf{y}'$ ensures $\Delta_1^2 + \Delta_2^2 > 0$, and all inventories are nonnegative. This establishes (21), yielding $F(\mathbf{y}, \mathbf{y}', \mathbf{c}) < 0$ and completing Case 3.

Case 4 ($y_1 + y_2 > 1$ and $y'_1 + y'_2 \leq 1$) is symmetric to Case 3 and follows by an analogous argument, completing the proof of strict monotonicity. $\square$

The monotonicity proof relied on two algebraic identities, which we now establish rigorously.

Lemma C.1 (Algebraic Identity). *Given two pairs of variables y_1, y_2 and y'_1, y'_2 with differences $\Delta_i := y_i - y'_i$, $i = 1, 2$. Let $\phi_r(x) := rx - \frac{1}{2}x^2$. The following identity holds:*

$$\sum_{i=1}^2 (\phi_{1-y_j}(y_i) - \phi_{1-y'_j}(y'_i)) \Delta_i = \Delta_1^2 + \Delta_2^2 - \frac{y_1 + y_2 + y'_1 + y'_2}{4} ((\Delta_1 + \Delta_2)^2 + \Delta_1^2 + \Delta_2^2). \quad (23)$$

Proof. Expanding the left side of (23) yields

$$\begin{aligned} & \sum_{i=1}^2 (\phi_{1-y_j}(y_i) - \phi_{1-y'_j}(y'_i)) \Delta_i \\ &= \left((1-y_2)y_1 - \frac{1}{2}y_1^2 - (1-y'_2)y'_1 + \frac{1}{2}(y'_1)^2 \right) \Delta_1 + \left((1-y_1)y_2 - \frac{1}{2}y_2^2 - (1-y'_1)y'_2 + \frac{1}{2}(y'_2)^2 \right) \Delta_2 \\ &= \left((y_1 - y'_1) - (y_1y_2 - y'_1y'_2) - \frac{1}{2}(y_1^2 - (y'_1)^2) \right) \Delta_1 + \left((y_2 - y'_2) - (y_1y_2 - y'_1y'_2) - \frac{1}{2}(y_2^2 - (y'_2)^2) \right) \Delta_2 \\ &= \left(\Delta_1 - (y_1y_2 - y'_1y'_2) - \frac{1}{2}\Delta_1(y_1 + y'_1) \right) \Delta_1 + \left(\Delta_2 - (y_1y_2 - y'_1y'_2) - \frac{1}{2}\Delta_2(y_2 + y'_2) \right) \Delta_2 \\ &= \Delta_1^2 + \Delta_2^2 - (y_1y_2 - y'_1y'_2)(\Delta_1 + \Delta_2) - \frac{1}{2}(\Delta_1^2(y_1 + y'_1) + \Delta_2^2(y_2 + y'_2)) \end{aligned}$$

Using $y_i = y'_i + \Delta_i$, we obtain the auxiliary identities

$$y_1y_2 - y'_1y'_2 = y'_1\Delta_2 + y'_2\Delta_1 + \Delta_1\Delta_2, \quad (24)$$

$$\Delta_i^2(y_i + y'_i) = (2y'_i + \Delta_i)\Delta_i^2, \quad (25)$$

$$\Delta_1 + \Delta_2 = (y_1 + y_2) - (y'_1 + y'_2), \quad (26)$$

$$\Delta_1^2 + \Delta_2^2 + \Delta_1\Delta_2 = \frac{1}{2}((\Delta_1 + \Delta_2)^2 + \Delta_1^2 + \Delta_2^2). \quad (27)$$

Applying these identities successively simplifies the remaining terms:

$$\begin{aligned} & (y_1y_2 - y'_1y'_2)(\Delta_1 + \Delta_2) + \frac{1}{2}(\Delta_1^2(y_1 + y'_1) + \Delta_2^2(y_2 + y'_2)) \\ & \stackrel{(24),(25)}{=} (y'_1\Delta_2 + y'_2\Delta_1 + \Delta_1\Delta_2)(\Delta_1 + \Delta_2) + \frac{1}{2}((2y'_1 + \Delta_1)\Delta_1^2 + (2y'_2 + \Delta_2)\Delta_2^2) \\ &= (y'_1\Delta_1\Delta_2 + y'_1\Delta_2^2 + y'_2\Delta_1\Delta_2 + y'_2\Delta_1^2) + \Delta_1\Delta_2(\Delta_1 + \Delta_2) + \frac{1}{2}(2y'_1\Delta_1^2 + 2y'_2\Delta_2^2 + \Delta_1^3 + \Delta_2^3) \\ &= (y'_1 + y'_2)(\Delta_1^2 + \Delta_2^2 + \Delta_1\Delta_2) + \frac{1}{2}(\Delta_1^3 + \Delta_2^3 + 2\Delta_1\Delta_2^2 + 2\Delta_1^2\Delta_2) \\ &= (y'_1 + y'_2)(\Delta_1^2 + \Delta_2^2 + \Delta_1\Delta_2) + \frac{1}{2}(\Delta_1 + \Delta_2)(\Delta_1^2 + \Delta_2^2 + \Delta_1\Delta_2) \\ & \stackrel{(26)}{=} \frac{y_1 + y_2 + y'_1 + y'_2}{2}(\Delta_1^2 + \Delta_2^2 + \Delta_1\Delta_2) \\ & \stackrel{(27)}{=} \frac{y_1 + y_2 + y'_1 + y'_2}{4}((\Delta_1 + \Delta_2)^2 + \Delta_1^2 + \Delta_2^2). \end{aligned}$$

Combining with the initial expansion establishes (23). $\square$

Lemma C.2 (Algebraic identity for mixed regimes). *Let $\mathbf{y} = (y_1, y_2)$ and $\mathbf{y}' = (y'_1, y'_2)$ with differences $\Delta_i := y_i - y'_i$, $i = 1, 2$. Define*

$$B_i := y_i \left(1 - y_j - \frac{y_i}{2} \right) - \frac{1}{2} (1 - y'_j)^2,$$

then

$$B_1 \Delta_1 + B_2 \Delta_2 = \Delta_1^2 + \Delta_2^2 - \frac{y_1 + y_2 + y'_1 + y'_2}{4} \left((\Delta_1 + \Delta_2)^2 + \Delta_1^2 + \Delta_2^2 \right) - \frac{(y'_1 + y'_2 - 1)^2}{2} (\Delta_1 + \Delta_2). \quad (28)$$

Proof. Introduce aggregate notation:

$$S := y_1 + y_2, \quad S' := y'_1 + y'_2, \quad \Delta_S := \Delta_1 + \Delta_2 = S - S'.$$

Expanding the left side of (28) gives

$$\begin{aligned} B_1 \Delta_1 + B_2 \Delta_2 &= \left(y_1 - y_1 y_2 - \frac{1}{2} y_1^2 - \frac{1}{2} (1 - y'_2)^2 \right) \Delta_1 + \left(y_2 - y_1 y_2 - \frac{1}{2} y_2^2 - \frac{1}{2} (1 - y'_1)^2 \right) \Delta_2 \\ &= \underbrace{(y_1 \Delta_1 + y_2 \Delta_2)}_{(L)} - \underbrace{y_1 y_2 (\Delta_1 + \Delta_2)}_{(M)} - \underbrace{\frac{1}{2} (y_1^2 \Delta_1 + y_2^2 \Delta_2)}_{(N)} - \underbrace{\frac{1}{2} ((1 - y'_2)^2 \Delta_1 + (1 - y'_1)^2 \Delta_2)}_{(O)}. \end{aligned}$$

We will expand each of the four terms (L)–(O) separately.

(L): Using $y_i = y'_i + \Delta_i$,

$$\begin{aligned} (L) &= y_1 \Delta_1 + y_2 \Delta_2 = (y'_1 + \Delta_1) \Delta_1 + (y'_2 + \Delta_2) \Delta_2 \\ &= y'_1 \Delta_1 + y'_2 \Delta_2 + \Delta_1^2 + \Delta_2^2. \end{aligned}$$

(M): Expanding the product gives

$$y_1 y_2 = (y'_1 + \Delta_1)(y'_2 + \Delta_2) = y'_1 y'_2 + y'_1 \Delta_2 + y'_2 \Delta_1 + \Delta_1 \Delta_2,$$

we have

$$\begin{aligned} (M) &= y_1 y_2 (\Delta_1 + \Delta_2) = (y'_1 y'_2 + y'_1 \Delta_2 + y'_2 \Delta_1 + \Delta_1 \Delta_2) \Delta_S \\ &= y'_1 y'_2 \Delta_S + \underbrace{(y'_1 \Delta_2 + y'_2 \Delta_1) \Delta_S}_{(M_1)} + \underbrace{\Delta_1 \Delta_2 \Delta_S}_{(M_2)}. \end{aligned}$$

Expanding (M_1) and (M_2) in terms of Δ_1, Δ_2 :

$$\begin{aligned} (M_1) &= (y'_1 \Delta_2 + y'_2 \Delta_1) (\Delta_1 + \Delta_2) = y'_1 \Delta_1 \Delta_2 + y'_1 \Delta_2^2 + y'_2 \Delta_1^2 + y'_2 \Delta_1 \Delta_2 \\ &= y'_2 \Delta_1^2 + y'_1 \Delta_2^2 + (y'_1 + y'_2) \Delta_1 \Delta_2 \\ &= y'_2 \Delta_1^2 + y'_1 \Delta_2^2 + S' \Delta_1 \Delta_2. \end{aligned}$$

and

$$(M_2) = \Delta_1 \Delta_2 (\Delta_1 + \Delta_2) = \Delta_1^2 \Delta_2 + \Delta_1 \Delta_2^2.$$

This gives

$$(M) = y'_1 y'_2 \Delta_S + y'_2 \Delta_1^2 + y'_1 \Delta_2^2 + S' \Delta_1 \Delta_2 + \Delta_1^2 \Delta_2 + \Delta_1 \Delta_2^2.$$

(N): Using $y_i^2 = (y'_i + \Delta_i)^2 = y_i'^2 + 2y'_i \Delta_i + \Delta_i^2$, we have:

$$\begin{aligned} (N) &= \frac{1}{2} (y_1^2 \Delta_1 + y_2^2 \Delta_2) = \frac{1}{2} \left((y_1'^2 + 2y'_1 \Delta_1 + \Delta_1^2) \Delta_1 + (y_2'^2 + 2y'_2 \Delta_2 + \Delta_2^2) \Delta_2 \right) \\ &= \frac{1}{2} (y_1'^2 \Delta_1 + y_2'^2 \Delta_2) + y'_1 \Delta_1^2 + y'_2 \Delta_2^2 + \frac{1}{2} (\Delta_1^3 + \Delta_2^3). \end{aligned}$$

(O): Use $(1 - y'_i)^2 = 1 - 2y'_i + y'^2_i$ to expand:

$$\begin{aligned} (O) &= \frac{1}{2}((1 - y'_2)^2 \Delta_1 + (1 - y'_1)^2 \Delta_2) = \frac{1}{2}((1 - 2y'_2 + y'^2_2) \Delta_1 + (1 - 2y'_1 + y'^2_1) \Delta_2) \\ &= \frac{1}{2}(\Delta_1 + \Delta_2) - (y'_2 \Delta_1 + y'_1 \Delta_2) + \frac{1}{2}(y'^2_2 \Delta_1 + y'^2_1 \Delta_2) \\ &= \frac{1}{2} \Delta_S - (y'_2 \Delta_1 + y'_1 \Delta_2) + \frac{1}{2}(y'^2_2 \Delta_1 + y'^2_1 \Delta_2). \end{aligned}$$

Combining all four terms yields

$$\begin{aligned} B_1 \Delta_1 + B_2 \Delta_2 &= (L) - (M) - (N) - (O) \\ &= (y'_1 \Delta_1 + y'_2 \Delta_2 + \Delta_1^2 + \Delta_2^2) - (y'_1 y'_2 \Delta_S + y'_2 \Delta_1^2 + y'_1 \Delta_2^2 + S' \Delta_1 \Delta_2 + \Delta_1^2 \Delta_2 + \Delta_1 \Delta_2^2) \\ &\quad - \left(\frac{1}{2}(y'^2_1 \Delta_1 + y'^2_2 \Delta_2) + y'_1 \Delta_1^2 + y'_2 \Delta_2^2 + \frac{1}{2}(\Delta_1^3 + \Delta_2^3) \right) \\ &\quad - \left(\frac{1}{2} \Delta_S - (y'_2 \Delta_1 + y'_1 \Delta_2) + \frac{1}{2}(y'^2_2 \Delta_1 + y'^2_1 \Delta_2) \right). \end{aligned}$$

Decompose $B_1 \Delta_1 + B_2 \Delta_2 = Q_1 + Q_2$, where Q_1 collects linear terms and Q_2 collects higher-order terms:

$$\begin{aligned} Q_1 &:= (y'_1 \Delta_1 + y'_2 \Delta_2) - (y'_1 y'_2 \Delta_S) - \frac{1}{2}(y'^2_1 \Delta_1 + y'^2_2 \Delta_2) - \frac{1}{2} \Delta_S + (y'_2 \Delta_1 + y'_1 \Delta_2) - \frac{1}{2}(y'^2_2 \Delta_1 + y'^2_1 \Delta_2), \\ Q_2 &:= \Delta_1^2 + \Delta_2^2 - (y'_2 \Delta_1^2 + y'_1 \Delta_2^2 + S' \Delta_1 \Delta_2 + \Delta_1^2 \Delta_2 + \Delta_1 \Delta_2^2) - (y'_1 \Delta_1^2 + y'_2 \Delta_2^2 + \frac{1}{2}(\Delta_1^3 + \Delta_2^3)). \end{aligned}$$

Simplifying Q_1 using $S'^2 = y'^2_1 + y'^2_2 + 2y'_1 y'_2$:

$$\begin{aligned} Q_1 &= \left(y'_1 + y'_2 - \frac{1}{2}(y'^2_1 + y'^2_2) - y'_1 y'_2 - \frac{1}{2} \right) (\Delta_1 + \Delta_2) \\ &= \left(S' - \frac{1}{2}(S'^2) - \frac{1}{2} \right) \Delta_S \\ &= -\frac{1}{2}(S' - 1)^2 \Delta_S. \end{aligned}$$

Simplifying Q_2 :

$$\begin{aligned} Q_2 &= \Delta_1^2 + \Delta_2^2 - (y'_2 \Delta_1^2 + y'_1 \Delta_2^2 + S' \Delta_1 \Delta_2 + \Delta_1^2 \Delta_2 + \Delta_1 \Delta_2^2) - (y'_1 \Delta_1^2 + y'_2 \Delta_2^2 + \frac{1}{2}(\Delta_1^3 + \Delta_2^3)) \\ &= (1 - y'_1 - y'_2)(\Delta_1^2 + \Delta_2^2) - S' \Delta_1 \Delta_2 - (\Delta_1^2 \Delta_2 + \Delta_1 \Delta_2^2 + \frac{1}{2}(\Delta_1^3 + \Delta_2^3)). \end{aligned}$$

The cubic terms factor as

$$\Delta_1^2 \Delta_2 + \Delta_1 \Delta_2^2 + \frac{1}{2}(\Delta_1^3 + \Delta_2^3) = \frac{1}{2} \Delta_S (\Delta_1^2 + \Delta_2^2 + \Delta_1 \Delta_2),$$

which reduces Q_2 to

$$Q_2 = (\Delta_1^2 + \Delta_2^2) - S'(\Delta_1^2 + \Delta_2^2 + \Delta_1 \Delta_2) - \frac{1}{2} \Delta_S (\Delta_1^2 + \Delta_2^2 + \Delta_1 \Delta_2).$$

Using $S = S' + \Delta_S$, we obtain the identities

$$\frac{S + S'}{4} = \frac{2S' + \Delta_S}{4} = \frac{S'}{2} + \frac{\Delta_S}{4}.$$

together with a second identity

$$\Delta_S^2 + \Delta_1^2 + \Delta_2^2 = (\Delta_1 + \Delta_2)^2 + \Delta_1^2 + \Delta_2^2 = 2(\Delta_1^2 + \Delta_2^2 + \Delta_1 \Delta_2).$$

Applying these identities gives

$$\begin{aligned} -S'(\Delta_1^2 + \Delta_2^2 + \Delta_1\Delta_2) - \frac{1}{2}\Delta_S(\Delta_1^2 + \Delta_2^2 + \Delta_1\Delta_2) &= -\left(\frac{S'}{2} + \frac{\Delta_S}{4}\right) \cdot 2(\Delta_1^2 + \Delta_2^2 + \Delta_1\Delta_2) \\ &= -\frac{S+S'}{4}(\Delta_S^2 + \Delta_1^2 + \Delta_2^2) \end{aligned}$$

which yields

$$Q_2 = \Delta_1^2 + \Delta_2^2 - \frac{S+S'}{4}(\Delta_S^2 + \Delta_1^2 + \Delta_2^2).$$

Combining Q_1 and Q_2 establishes (28):

$$B_1\Delta_1 + B_2\Delta_2 = Q_2 + Q_1 = \Delta_1^2 + \Delta_2^2 - \frac{S+S'}{4}(\Delta_S^2 + \Delta_1^2 + \Delta_2^2) - \frac{(S'-1)^2}{2}\Delta_S.$$

□

Table 1. Complete information game: Deviation of learned action from best action (Nash equilibrium/Best Response) in symmetric and asymmetric settings after 100 iterations.

Scenario	Agent	Action	Std	BR/NE	BR Std	Loss Abs.	Rel. (%)
Symmetric	1	0.586135	0.008301	0.585786		0.000349	0.060
	2	0.588358	0.008516	0.585786		0.002572	0.439
Asymmetric	1	0.807032	0.010333	0.804594	0.008614	0.002438	0.303
	2	0.219723	0.003808	0.219061	0.004055	0.000662	0.302
Asymmetric $p = 0.2$	1	0.755007	0.007003	0.754350	0.005375	0.000657	0.087
	2	0.358770	0.003084	0.358065	0.003785	0.000705	0.197
Asymmetric $q = 0.2$	1	0.905380	0.010280	0.904399	0.003329	0.000981	0.108
	2	0.255479	0.002889	0.252230	0.003455	0.000249	0.098
Asymmetric $p=q=0.2$	1	0.837988	0.007729	0.839101	0.003329	0.001112	0.133
	2	0.411364	0.004047	0.409971	0.003455	0.001394	0.340

D. Experimental Setup and Hyperparameters

We employ Proximal Policy Optimization (PPO) for all deep reinforcement learning experiments due to its empirical stability and scalability to continuous action spaces. Our implementation utilizes fully connected neural networks with two hidden layers, each containing 64 nodes with SeLU activation functions. The network weights and biases constitute the policy parameters θ_i for each agent i .

All experiments are conducted using parallel environment simulation with a batch size of 20,000 steps per policy update iteration. We employ the Adam optimizer with a learning rate of 10^{-5} across all experimental scenarios. The PPO-specific hyperparameters include a clipping parameter $\epsilon = 0.2$ and entropy coefficient of 0.01 to encourage exploration during training.

State representation. In all runs, each agent observes only its own private cost type, encoded as a single scalar in $[0, 1]$ (types are discretized in that interval); no opponent or global features are observed. Actions are sampled from the policy over the discretized action grid used for verification.

Verification procedure. We employ two different verification procedures. In the symmetric complete-information game we have with Theorem 4.2 a closed-form expression for the unique Nash equilibrium, which we use for verification. In all other settings, we compute the best response to the learned policy using numerical integration over the discretized action space and type distribution.

Best-response verification. For each cost type, the action space is discretized into 1024 grid points. We simulate 12 million episodes against the learned policies to compute the expected best response and the deviations reported in the tables.

For robustness validation, we evaluate each experimental configuration across 10 independent random seeds (0–9) and report averaged results; the standard deviation across seeds is consistently low, as can be seen in the Tables 1 and 2. Complete-information game experiments converge within 20–50 iterations; to ensure stability, we run all such experiments for 100 iterations. Bayesian game experiments typically require more iterations due to the added complexity of type distributions and strategic uncertainty, thus, we run these games for 400 iterations.

For equilibrium verification, we employ the multi-agent equilibrium verification framework adapted from Pieroth et al. (2025), using a discretization parameter of $D = 1024$ and an 12 million simulations. The complete hyperparameter configuration is summarized in Table 3 below. However, it should be noted that we did not perform an exhaustive hyperparameter search. The chosen hyperparameters are the standard values used in the default setting from Pieroth et al. (2025). We only ran an additional sensitivity analysis focused on the learning rate, which is commonly considered the most influential hyperparameter, and found that convergence is robust across a broad range (10^{-3} to 10^{-6}) of learning rates, with 10^{-5} providing a good balance between convergence speed and stability in our experiments.

All computations are performed on a single Nvidia GeForce 2080Ti GPU with 11 gigabytes of RAM, with each experimental run requiring approximately from 2 minutes to an hour of computation time depending on the scenario complexity, convergence requirements, and mostly frequency of evaluation during training.

Table 2. Bayesian game: Deviation of learned action from best response in symmetric and asymmetric (Uniform and Gaussian) settings with different cost types.

Scenario	Agent	Cost	Prob	Action	Action Std	Best Resp.	BR Std	Loss Abs.	Rel. (%)
Symm. Uniform	1	0.3	40 %	0.816919	0.005138	0.818377	0.007326	0.001458	0.178
		0.7	60 %	0.402475	0.007081	0.402542	0.004762	0.000067	0.017
	2	0.3	40 %	0.813841	0.004366	0.814663	0.007836	0.000821	0.101
		0.7	60 %	0.400402	0.005900	0.402346	0.004838	0.001944	0.483
Symm. Gaussian	1	0.3	40 %	0.643952	0.002922	0.640567	0.003938	0.003386	0.529
		0.7	60 %	0.459024	0.001841	0.461291	0.002665	0.002267	0.491
	2	0.3	40 %	0.643262	0.002203	0.640469	0.003709	0.002792	0.436
		0.7	60 %	0.458989	0.002445	0.461388	0.002210	0.002399	0.520
Asymm. Uniform	1	0.3	40 %	0.833658	0.005907	0.829032	0.005671	0.004626	0.558
		0.7	60 %	0.406724	0.006825	0.406745	0.003840	0.000021	0.005
	2	0.2	50 %	0.914358	0.006822	0.912024	0.007430	0.002334	0.256
		0.8	50 %	0.280108	0.005470	0.280156	0.004253	0.000048	0.017
Asymm. Gaussian	1	0.3	40 %	0.652818	0.002490	0.647116	0.002726	0.005702	0.881
		0.7	60 %	0.462185	0.001837	0.463930	0.001796	0.001745	0.376
	2	0.2	50 %	0.699764	0.003095	0.695015	0.002726	0.004750	0.683
		0.8	50 %	0.404407	0.002904	0.407331	0.002397	0.002924	0.718

Table 3. Hyperparameters used for PPO.

Parameter	Value
Algorithm	PPO (Clipped)
Hidden Layers	2
Hidden Units per Layer	64
Activation function	SeLU
Learning Rate	10^{-5}
Discount factor (γ)	1.0
Batch Size	20k
Clip Parameter (ϵ)	0.2
Entropy Coefficient	0.01
Optimizer	Adam
Training Iterations	100 to 400
Number of seeds	10 (0 to 9)
Action discretization	1024
Simulations for verification	12 million

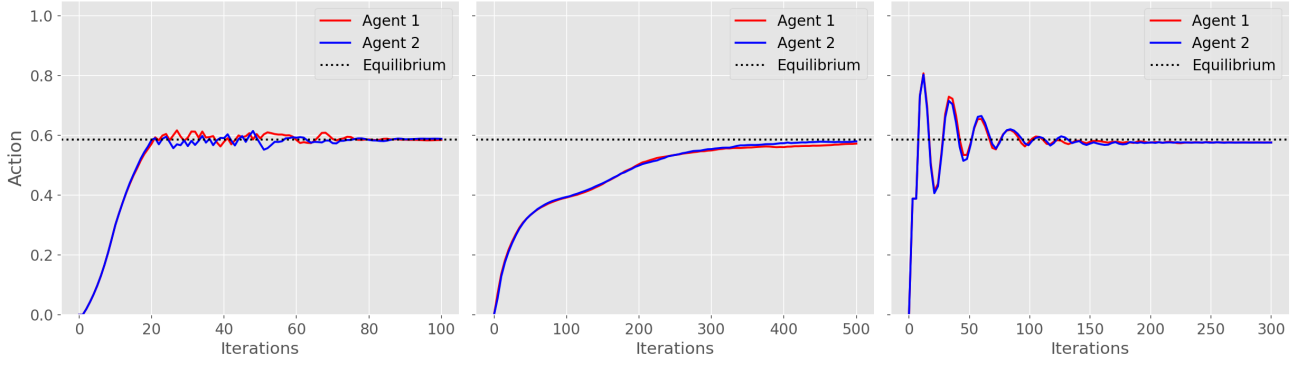


Figure 2. Convergence of agent actions in the symmetric complete-information game across algorithms: PPO (left), Reinforce (middle), and TD3 (right), benchmarked against the Nash equilibrium (NE).

Table 4. Symmetric complete information game: Deviation of learned action from Nash equilibrium across different algorithms (all results reported after 600 training iterations).

Algorithm	Agent	Action	Std	NE	Loss Abs.	Rel. (%)
ppo	1	0.582377	0.017259	0.585786	0.003409	0.5819
	2	0.593071	0.020221	0.585786	0.007285	1.2436
reinforce	1	0.583129	0.005484	0.585786	0.002657	0.4535
	2	0.584431	0.002666	0.585786	0.001355	0.2313
td3	1	0.587717	0.018602	0.585786	0.001931	0.3296
	2	0.589470	0.013331	0.585786	0.003684	0.6289

E. Comparison of Learning Algorithms

We use PPO as a representative off-the-shelf deep reinforcement learning method to test whether standard independent learners also converge in this structurally favorable game, even though our strict monotonicity result provides formal convergence guarantees only for variational-inequality methods such as extra-gradient and Mirror-Prox. However, there are many other algorithms that could be applied to this problem, and it is natural to ask whether the observed convergence behavior is specific to PPO or more general across different learning methods. To address this question, we compare PPO against two additional algorithms: REINFORCE, a classic policy gradient method, and TD3, a state-of-the-art actor-critic method for continuous control.

For the symmetric complete information game, the results can be seen in Figure 2 and Table 4, while for the asymmetric complete information game, the results are shown in Figure 3 and Table 5. In both cases, PPO, REINFORCE, and TD3 converge to the Nash equilibrium or best response with small deviations, however, it can be observed that the convergence of PPO is more stable and closer to the equilibrium compared to REINFORCE, which exhibits higher variance and larger deviations. TD3 also shows good convergence properties, but with slightly slower convergence than PPO. These results suggest that while multiple algorithms can achieve convergence in this setting, PPO’s stability and performance make it a particularly effective choice for learning in these games.

However, as with PPO, we did not perform an exhaustive hyperparameter search for each algorithm since the main purpose of this comparison is to demonstrate that the observed convergence behavior is not specific to PPO, rather than to identify the best possible performance for each algorithm. With further tuning, it is possible that with different hyperparameters, REINFORCE and TD3 could achieve faster convergence properties.



Figure 3. Convergence of agent actions in the complete-information game across algorithms: PPO, Reinforce, and TD3, compared against the best response (BR).

Table 5. Deviations from Nash equilibrium or best response across algorithms for the asymmetric complete-information game and the Bayesian game in symmetric and asymmetric settings. All experiments use 1200 iterations.

Scenario	Algor.	Agent	Cost	Action	Std	BR/NE	Loss Abs.	Rel. (%)
CIG Asymmetric	ppo	1	0.5	0.798023	0.015595	0.803519	0.005496	0.6804
		2	0.8	0.222389	0.007049	0.221408	0.000982	0.4433
	reinforce	1	0.5	0.790820	0.006392	0.801173	0.010353	1.2921
		2	0.8	0.223738	0.002589	0.222678	0.001045	0.4676
	td3	1	0.5	0.813621	0.015825	0.806354	0.007267	0.9011
		2	0.8	0.218195	0.010739	0.217889	0.000307	0.1404
Bayesian Symmetric	ppo	1	0.3	0.813734	0.014725	0.813783	0.000049	0.0060
			0.7	0.397297	0.010526	0.400684	0.003387	0.8454
		2	0.3	0.810604	0.013632	0.814760	0.004157	0.5102
			0.7	0.404945	0.016420	0.400684	0.004260	1.0633
	reinforce	1	0.3	0.807639	0.008664	0.809971	0.002332	0.2879
			0.7	0.401338	0.004408	0.402151	0.000813	0.2021
		2	0.3	0.811303	0.010269	0.819355	0.008052	0.9828
			0.7	0.402525	0.006470	0.403226	0.000701	0.1739
	td3	1	0.3	0.816881	0.020566	0.813294	0.003586	0.4410
			0.7	0.404212	0.018270	0.400586	0.003626	0.9051
		2	0.3	0.820480	0.014665	0.812805	0.007674	0.9442
			0.7	0.403482	0.015068	0.401368	0.002114	0.5267
Bayesian Asymmetric	ppo	1	0.3	0.831968	0.011010	0.834311	0.002343	0.2808
			0.7	0.404158	0.016756	0.407038	0.002880	0.7076
		2	0.2	0.911933	0.016225	0.914370	0.002437	0.2665
			0.8	0.281657	0.020338	0.282405	0.000748	0.2647
	reinforce	1	0.3	0.820977	0.008528	0.833529	0.012552	1.5059
			0.7	0.403989	0.003735	0.404595	0.000606	0.1497
		2	0.2	0.899734	0.009189	0.910362	0.010628	1.1675
			0.8	0.288216	0.004981	0.286315	0.001901	0.6641
	td3	1	0.3	0.835938	0.006097	0.833431	0.002507	0.3008
			0.7	0.409409	0.011846	0.406158	0.003251	0.8003
		2	0.2	0.913775	0.009955	0.910850	0.002925	0.3211
			0.8	0.278359	0.013428	0.280645	0.002286	0.8144

Table 6. Hyperparameters used for PPO, REINFORCE, and TD3. Parameters not listed here are either explicitly specified in the comparison or the same as in the baseline configuration (Table 3)

Algorithm	PPO (Clipped)	REINFORCE	TD3
Learning Rate	10^{-5}	10^{-2}	5×10^{-4}
Learning Rate Scheduler	multistep	multistep	linear
Clip Parameter (ϵ)	1.0	-	-
GAE λ	0.95	-	-
Full Batch Updates	true	-	-
Entropy Coefficient	0.01	-	-
Gradient Steps	-	-	2
Replay Buffer Size	-	-	400k
Learning Starts	-	-	400k
Target Update (τ)	-	-	1.0
Policy Delay	-	-	2
Target Policy Noise	-	-	0.2
Noise Clip	-	-	0.5
Action Noise	-	-	$\mathcal{N}(0, 0.5)$ (linear decay)

F. Notation and Symbols

This section summarizes all symbols used throughout the paper for ease of reference.

F.1. Sets and Indices

- i, j : Vendor (player) indices, $i \in \{1, 2\}$, $j = 3 - i$.
- $K_i \subseteq [0, 1]$: Action space (inventory levels) of vendor i .
- $K := K_1 \times K_2$: Joint action space (complete-information game).
- $\mathcal{K}_i := K_i^{|C_i|}$: Strategy space of vendor i (Bayesian game).
- $\mathcal{K} := \mathcal{K}_1 \times \mathcal{K}_2$: Joint strategy space (Bayesian game).
- $C_i \subset [0, 1]$: Finite type (cost) space of vendor i .
- $C := C_1 \times C_2$: Joint cost space.
- $\mathcal{C}$: Set of admissible cost vectors in the complete-information game.
- L_i : Cost lottery for vendor i in Bayesian scenarios.

F.2. Decision Variables and Strategies

- y_i : Inventory level chosen by vendor i .
- $y = (y_1, y_2)$: Joint inventory vector.
- $y_i(\cdot)$: Strategy of vendor i in the Bayesian game, mapping types to actions.
- y^* : Nash equilibrium inventory vector.
- $\Delta_i := y_i - y'_i$: Difference in inventory levels.
- $\Delta := y - y'$: Joint difference vector.

F.3. Demand and Random Variables

- x_i : Baseline stochastic demand faced by vendor i .
- $f_i(\cdot)$: Probability density function of x_i .
- $f(\cdot)$: Probability density function of agent i .
- $g(\cdot)$: Probability density function of rival's (agent j) demand when agent i is clear from context.
- $F_i(\cdot)$: Cumulative distribution function of x_i .
- d_i : Spillover demand faced by vendor i from vendor j .
- B_i : Spillover threshold determining redirected demand, $B_i(x_i) = \frac{y_i - x_i}{a} + y_j$.
- $(\cdot)^+$: Positive-part operator, $(z)^+ = \max\{z, 0\}$.

F.4. Economic Parameters

- c_i : Unit inventory cost of vendor i .
- s : Selling price.
- p : Penalty cost for unmet original demand.
- q : Salvage value for unsold inventory.
- $a \in [0, 1]$: Substitution (spillover) parameter.

F.5. Profit and Payoff Functions

- $\pi_i(y, c_i, x)$: Realized profit of vendor i .
- $J_i(y, c_i)$: Expected profit of vendor i .
- $J(y, c) := (J_1(y, c_1), J_2(y, c_2))$: Vector of expected profits.
- $I_i(y, c_i) := \partial J_i(y, c_i) / \partial y_i$: Marginal profit in the complete-information game.
- $I(y, c)$: Vector of marginal profits.
- $\hat{J}_i(y, c_i)$: Bayesian expected profit conditional on type c_i .
- $\hat{I}_i(y, c_i)$: Bayesian marginal profit operator.
- $\hat{I}(y, c)$: Bayesian marginal profit vector.
- $\bar{I}(y) := \mathbb{E}_c[\hat{I}(y, c)]$: Ex-ante expected marginal profit operator.

F.6. Equilibrium Concepts

- NE : Nash equilibrium.
- BNE : Bayes–Nash equilibrium.
- BR : Best response.

F.7. Variational Inequality and Monotonicity

- $F(y) := -\nabla J(y)$: Variational inequality operator.
- $\langle \cdot, \cdot \rangle$: Euclidean inner product.
- $\nabla J(y)$: Gradient of the payoff vector (Jacobian matrix).

F.8. Learning Algorithms and Numerical Parameters

- PPO : Proximal Policy Optimization algorithm.
- ϵ : PPO clipping parameter.
- θ_i : Policy parameters of agent i .